

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)			Red blood cell / RBC correctly labelled White blood cell / WBC correctly labelled Both correct for 1 mark Ignore direction of arrow. Arrow must touch cells	1			1		
	(b)	(i)		3 marks 57 2 marks for sight of 57.0877944 / 57.33333333 (if rounded earlier) $\frac{2666}{4670} \times 100$ $\frac{43}{75.3225806} \times 100$ 1mark for sight of 43 x 62 = 2 666 4670/62 = 75.3225806		3		3	3	
		(ii)		{Transport / carries} {CO ₂ / soluble food / urea / hormones / antibodies / glucose / amino acids / vitamins / minerals / fats/ products of digestion} / distribution of heat (1) Ignore: carry {blood cells / nutrients}	1			1		

	(c)	(i)	Any two (x1) from: • least red blood cells / RBC below normal range / RBC lower than other patients / less RBC than other patients / use of data, e.g. RBC is 4.4 and the range is 4.5 - 5.2 (1) • Least haemoglobin / haemoglobin below normal range / haemoglobin lower than other patients / less haemoglobin than other patients / use of data, e.g. haemoglobin is 10.9 and the range is 13.2 - 14.8 (1) • cannot {carry / transport} enough oxygen / {red blood cells / haemoglobin} {carry / transport} oxygen (1)	1	1		2		
		(ii)	• {least / lowest concentration of / lowest number of} white blood cells / less white blood cells than the other patients (must be comparative) (1) • {defend / protects} against {disease / pathogens / bacteria / fungi / viruses / microbes / microorganisms} / {Kill / destroy / engulf} {bacteria / fungi / viruses / microbes / microorganisms / pathogens} / correct ref to antibody production(1) Ignore references to infection	1	1		2		
			Question 3 total	4	5	0	9	3	0